Table pK_b's of substituted anilines (pK_b value of aniline is 9.376)

Substituent	pK_b	Substituent	pK_b	Substituent	pK_b
o - CH_3	9.60	m - CH_3	9.31	p - CH_3	8.92
o - NH_2	9.52	m - NH_2	9.00	P - NH_2	7.83
o - OCH_3	9.52	m - OCH_3	9.70	p - OCH_3	8.70
o - NO_2	14.30	m - NO_2	11.52	p - NO_2	13.00
o - Cl	11.25	m - Cl	10.52	p - Cl	10.00

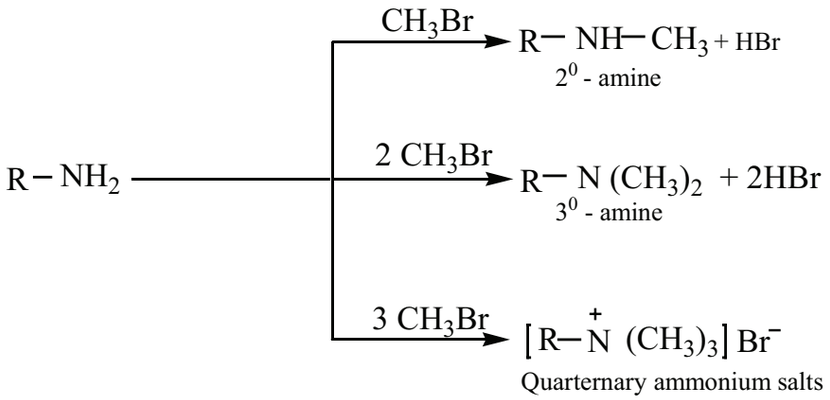
The relative basicity of amines follows the below mentioned order

Alkyl amines > Aralkyl amines > Ammonia > N – Aralkyl amines > Aryl amines

13.2.6 Chemical properties of amines

1) Alkylation

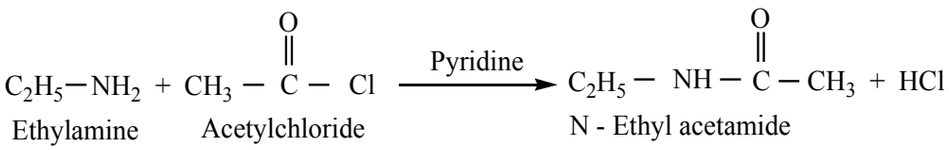
Amines reacts with alkyl halides to give successively 2^0 and 3^0 amines and quaternary ammonium salts.



2) Acylation

Aliphatic / aromatic primary and secondary amines react with acetyl chloride (or) acetic anhydride in presence of pyridine to form N – alkyl acetamide.

Example



3) Schotten – Baumann reaction

Aniline reacts with benzoylchloride (C_6H_5COCl) in the presence of NaOH to give N – phenyl benzamide. This reaction is known as Schotten – Baumann reaction. The acylation and benzoylation are nucleophilic substitutions.

XII U13-Organic Nitrogen Compounds.indd 214 16-12-2022 19:45:21